Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/31/2006
PI Name:	Overland, David M.S.		
Project Title:	Advanced Control System Development for Integration of Advanced Life Support Systems		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	NASA - Johnson Space Center		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	INTERNAL PROJECT
Start Date:	10/01/1998	**End Date:**	09/30/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Edeen, Marybeth		
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The purpose of this project is to develop advanced control system technology as part of an integrated approach to the command and control system of advanced life support. The control strategies for ALS systems must deal with continuous and discrete processes and with the dynamical interactions between sub-systems such as air revitalization, water recovery, food production, solids processing and the crew. The goal of autonomously controlling an ALS system challenges many areas of research, including software engineering, distributed data management and control, sensor interpretation, planning and scheduling, modeling and simulation, and validation and verification of autonomous control systems. These various research areas must be integrated into a coherent system that runs day after day for years and that can effectively interact with crewmembers, ground support and other systems. The control strategy must be able to reach “across” the system and “down” into its parts to gather information necessary to achieve mission objectives. Interfaces to crew, ground control, and other spacecraft systems must provide management of, and insight into		

	operations, while still supporting human tasks. The current task objective is to determine automation and autonomy technology gaps and risks, and to ultimately devise a roadmap for developing required technology and mitigating those risks. Advanced control system technology is a necessary discipline for NASA exploration missions, yet because it is also a new discipline, not previously required for human spaceflight, technology risks and gaps are not brought forward in exploration mission planning. Automation systems requirements engineering falls outside all currently funded activities.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work will benefit development of integrated control systems for large complex integrated systems where human intervention needs to be limited, but available on demand. Possible applications include remote processing plants or work in hostile environments. The technology also enables deeper integration of process goals and operations with continuous control and automation. This capability has great potential in the process control industry. A third benefit is the impact on software and system engineering processes as we learn to integrate large complex systems that also require complex advanced software.
Task Progress:	- The NASA Workshop on Advanced System Integration and Control for Life Support (ASICL) was held in August, 2003. Unfortunately, the Office of Biological and Physical Research (OBPR), Bioastronautics Research Division, has not yet issued the Workshop Final Report. - PI is continuing to actively participate as an Interdisciplinary Team member of the Advanced Integration Matrix (AIM) Program. Team Bravo was formed under AIM to investigate issues associated with advanced control and software systems integration. One released and several draft white papers have been generated as a result of this effort. - The FY04 AIM Benchmark Test is specifically designed to investigate advanced control architecture issues and technology gaps. The test is specifically designed to drive out a particular set of questions, which when answered in follow-on integration tests, will provide Evaluation Criteria for trade studies in control architectures, methods and automation technologies. The test is tentatively scheduled for early FY05. - A formal proposal was made to the Advanced Environmental Monitoring and Control (AEMC) Program for review in order to validate the legitimacy of the FY04 AIM Benchmark Test goals. - A NASA Summer Faculty Fellowship professor, Dr. Karlene Hoo from Texas Tech, was sponsored at JSC to support the process and controls design portion of the FY04 AIM Benchmark Test. - The paper “Control System Architectures, Technologies, and Concepts for Near Term and Future Human Exploration of Space” was presented at the Society for Automotive Engineers (SAE) 34th International Conference for Environmental Systems (ICES) in Colorado Springs, CO. This paper focused on design methodologies and architectures for the types of control systems required for exploration missions. - Small Business Innovative Research (SBIR) Subtopic B3.08 - Software Architectures and Integrated Control Strategies for Advanced Life Support Systems is reviewing Phase II proposals and is anticipating the Phase I proposal review. Subtopic goals were slightly modified to more closely align with the Office of Exploration goals. - Precursor aspects of this task, particularly the autonomy task definitions and automation technology surveys, were proposed in response to the Office of Exploration Systems Human and Robotic Technology Intramural Call. Proposal has been rejected.
Bibliography Type:	Description: (Last Updated: 06/19/2006)
Papers from Meeting Proceedings	Boulanger, Richard; Overland, David A. "Control System Architectures, Technologies, and Concepts for Near Term and Future Human Exploration of Space" SAE Paper 2004–01–2478, 34th ICES Jul-2004
Presentation	Overland, David; Ciskowski, Marvin; Cavanaugh, Ivan; Labuda, Laura; Mount, Frances; Whitmore, Mihriban; Gomez, Reynaldo; Slade, Howard; Park , John "Investigation of advanced control architecture issues and technology gaps" Advanced Integration Matrix Team Bravo White Paper - AIM Working Group 3 review Feb-2004